

Manual

Composition MPL-GAT 8.2

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1 Composition

Summary

General

Composition means the chemical make-up of input materials that are melted down in foundry to produce a defined alloy. Furthermore, composition makes sure (by calculation) that the chemical composition of the alloy can be achieved by adding as few ingredients as possible taking into account the current material stock. The objective is to use materials in a cost-effective manner in the production process.

Purpose

The function package "composition" supplements the function packages "shop floor data collection" and "material and production logistics" by functions to manage recipes, optimize material usage and analyze materials for the production of alloys.

Implementation Considerations

The function package "composition" is used if

- You would like to define and manage composition recipes for the production of alloys in the system.
- You would like to use the composition function
 - as an analysis tool to make sure by calculation that the chemical composition of the melt matches the target specifications of an alloy.
 - to plan material usage for the production of alloys taking into account the actual material stock.

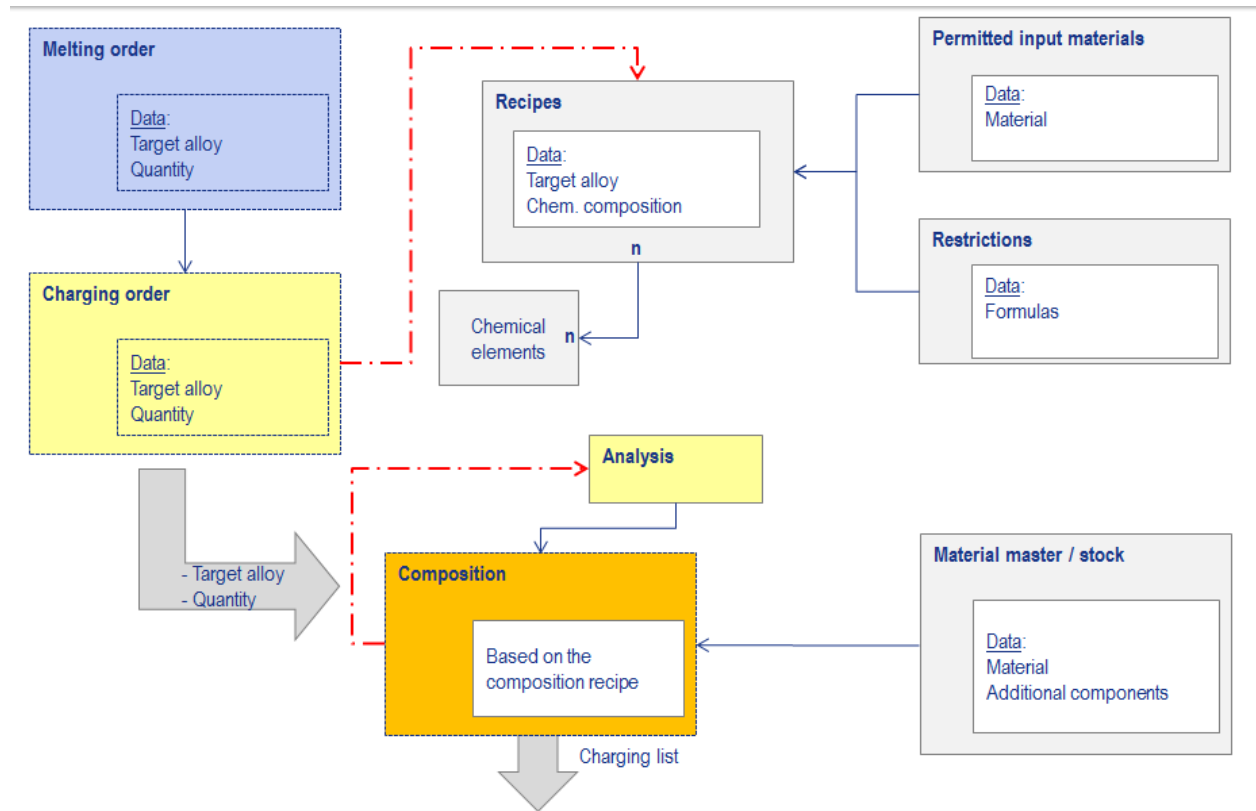
Features

- Configuration of the material master
- Configuration of the composition recipe
- Configuration of restricted material usage
- Configuration of permitted materials
- Composition/re-composition
- Charging process at the terminal
 - Log on order
 - Perform charging
 - Confirm charging
 - Sample taking

- Cast

Integration

The sections that follow describe the functional connection of individual components and functions necessary to perform composition in the system.



Orders/OP

The following orders/OPs are used in MES, in particular in connection with composition. They each have a special order type and differ from common production orders/OP.

- Melting order
Normally, "melting orders" are transferred from the ERP system to MES. They include the quantity as well as alloy to be produced.
- Charging order
If the melting order exists in MES, a "charging order" may be generated from it (composition --> generate charging order). Primarily, it has been copied from the melting order and takes over the relevant data.

The generated charging order is shown in the list of charging orders within the function "composition".

Chemical composition/composition recipe

The composition function shows the composition recipe in addition to the target alloy (article) of the charging order.

The composition recipe is defined for each target alloy within MES. Where it may be:

- edited in general
- fed with characteristics (chemical elements). The target value, upper tolerance limit as well as lower tolerance limit also have to be specified for these characteristics.
- released
 - Then the status of the composition recipe is "released" and no longer "in process". Only released composition recipes can be enabled/activated.
- enabled/disabled
 - If activated, a composition recipe will be used for the composition of an alloy or its sampling.

A composition recipe normally only includes chemical elements. Alloys from the material master are not used.

Permitted input materials and material usage restrictions

The function "permitted input materials" defines which input materials may at all be used for the composition of an alloy. This also identifies which materials (provided that material is available) are actually shown for an alloy within the "composition" function. Materials that are not permitted or permitted materials that are not available are not displayed.

All materials available will be shown if selection within the composition function is not restricted to the permitted input materials.

Several additional restrictions (formulas) may be defined by the function "material usage restriction". They will also be shown in the composition function. But restrictions can only be used optionally.

Material stock/ Material master

All materials used in composition have to be defined (anonymously, without batch reference) in the MES material master.

Defined materials are:

- Chemical elements: they are used as inspection characteristic within the composition recipe.
- Input materials: These can be:
 - Raw material
 - Alloys

- Scrap/recycling material (alloys)
- Target alloys

The MES keeps the inventory of materials defined in the material master.

2 Setup of Composition

Usage

You use the "composition" application to control planning, monitoring and execution of the melting process.

The sections that follow describe how to configure composition functions.

General

Composition always uses unique units of weight ("kg", "KG", "t"). Consequently, the target quantity unit of charging orders is identical to the unit of weight of input materials in the batch stock.

Basic configuration

Please proceed as follows if the composition function is installed on an already existing HYDRA 8 system:

1. Enable the patch dbp_mpl_composition as follows:
 - a. UNIX systems (run at server prompt within the HYDRA directory): *hydscri.out db_sql/dbp_mpl_composition.hsc*
 - b. Windows systems (run in a DOS window within the HYDRA directory): *hydscri.exe db_sql/dbp_mpl_composition.hsc*
2. Check the patch output
3. Save the existing dialog configuration:
 - a. UNIX systems (run at the server prompt in the HYDRA directory): *hydlgcfg.out DLGCFG.TYP=% DLGCFG.DLGUSR=% DLGCFG.DLG=% DATEI=dialog.dlg*
 - b. Windows systems (run in a DOS window within the HYDRA directory): *hydlgcfg.exe DLGCFG.TYP=% DLGCFG.DLGUSR=% DLGCFG.DLG=% DATEI=dialog.dlg*
4. Now load the new dialog configurations by the following command:
 - a. UNIX systems:
hymw.out -u9999 -b db_sql/aip_gat.dlg
 - b. Windows systems:
hymw.exe -u9999 -b db_sql/aip_gat.dlg

Now the dialogs C_CHPF, C_CHCF, C_CHTS, C_CHCA are imported **as template** (with type "DEF" and dialog user "999") or, if necessary, existing dialogs are updated.

5. Copy the dialogs C_CHPF, C_CHCF, C_CHTS, C_CHCA from the template (type "DEF" with dialog user "999") to type "DEF" and dialog user "0" using the MOC application: system administration --> terminals --> dynamic dialogs (transaction code ddconf). To do so, switch to the HYDRA Professional Mode.
6. Enable the new dynamic dialogs:

- a. UNIX systems:

hydialog.scr AIPTNR 0

- b. Windows systems:

sh.exe hydialog.scr AIPTNR 0

Please note: This command enables the default dialogs. Provided that dialogs specific to the terminal are used on the system, they need to be modified by an MPDV consultant.

Basic parameter settings

Enable the option "Automatic generation of batch no. when creating batches" in the MPL tab of the basic settings in MOC (system administration --> system settings --> basic settings).

Configuration: INI configuration

The below-mentioned settings entered in the [INI configuration](#) "MPL" within the section "COMPOSITION" affect calculation of the average material price for composition within the material master:

Key	Value	Comment
FIELD_SUM_CONSUMPTION	ist_lst_01	DB field for sum of consumption in order status (Sum of consumption)
FIELD_SUM_TOTAL_PRICE	ist_lst_02	DB field for sum of the total price in order status (Total price for all component entries)
CALCAVG_BEGIN_DAYS	10	Begin date (= today - <value>) for calculation of the average price
CALCAVG_END_DAYS	5	End date (= today - <value>) for calculation of the average price

The following conditions need to apply:

- CALCAVG_BEGIN_DAYS >= CALCAVG_END_DAYS >= 0
- Valid columns of the order status (table auftrag_status) have to be indicated for the fields FIELD_SUM_CONSUMPTION and FIELD_SUM_TOTAL_PRICE.

Please note:

- The configured columns (ist_lst_01 and ist_lst_02) are calculated when finishing the charging order.
- The ini configuration is not enabled by default.

Scheduler configuration

Enter the following values in MOC System administration → System settings → [Scheduler](#) in order to calculate average values or material costs and to complete inspection requests for finished orders:

Type	S – Standard
Alterable	Yes
Type	F - Fixed
Visible	Visible
Active	☒ Active
HYDRA user	0
Command	sh.exe hycompupd.scr (Windows) hycompupd.scr (Unix)
Comment	Composition tool
Fixed point in time	Hour: 0 Minute: 30

Configuration of units

Create the [unit](#) "%" (if it does not yet exist) to be able to use it for composition recipes.

Activation at the machine

Define the previous and subsequent material buffer in the [machine master](#) --> workplace configuration --> MPL:

Parameter name	Value
Workplace type	M = melting aggregate / furnace
Maximum capacity	Specification of the maximum load capacity of the furnace in kg
Batch management	N

Parameter name	Value
Preceding material buffer	Material buffer of the type C (casting buffer) This material buffer is the collection point in front of the furnace. Input materials are gathered on this material buffer. This material buffer needs to be assigned to <u>exactly</u> one machine (unique assignment). It is not allowed to assign it to several machines. But the system does not prevent it.
Subsequent material buffer	Material buffer of the type C (casting buffer) This material buffer represents the contents of the furnace. Once the charging process has been confirmed (i.e. the materials gathered on the collection point have been put into the furnace), this buffer includes a batch of the output material. This material buffer needs to be assigned to <u>exactly</u> one machine (unique assignment). It is not allowed to assign it to several machines. But the system does not prevent it.

Activation at the material buffer

Configure one or several [material buffers](#) as casting buffers that include the anonymous (input) material (see material type):

Parameter name	Value
Type	C =Casting buffer

Activation at the material type

Configure a [material type](#) for the batches included in the casting buffers:

Parameter name	Value
Input batch processing: inventory management	A = Anonymous

Definition of input materials

Create all materials that you would like to use as input material for composition in the [material master](#).

Parameter name	Value
Material number	"Material number"
Input material for composition: Input material	☒

Activation at order type

Configure an [order type](#) for charging orders. Scheduling must not release the order.

Parameter name	Value
Order type	e.g. "CHRG"
Options/order type for composition	C = charging order
Planning/scheduling without implicit release of orders	☒

Configure an order type for melting orders.

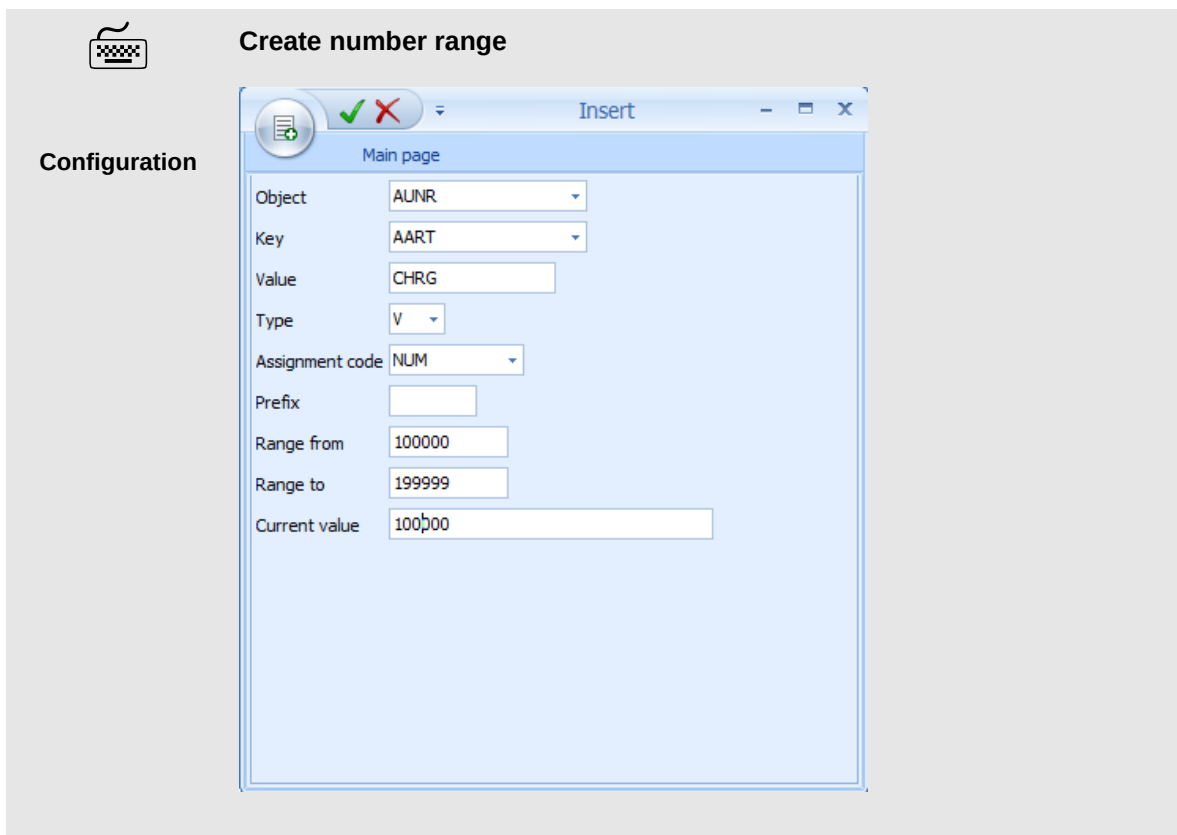
Parameter name	Value
Order type	e.g. "MELT"
Options/order type for composition	M = melting order

Please note that the generation of inspection requirements is pre-configured for the order type "CHRG" if the order status switches from "N" (not free) to "V" (prepared). Customizing services by MPDV are required if another order type is used (table: ade_aart_cbereich).

Configuration of number ranges

Create a [number range](#) for charging orders (object "AUNR", key "AART" and value "CHRG" or the relevant order type for charging orders). This number range is required to be able to create charging orders using the function "generate charging order" in the MOC application "composition".

Example



Create number range

Configuration

Object: AUNR

Key: AART

Value: CHRG

Type: V

Assignment code: NUM

Prefix:

Range from: 100000

Range to: 199999

Current value: 100000

Configuration of order statuses

Configure the [order status](#) "N" as the initial order status for the order type that has been defined as charging order by the order type for composition (e.g. "CHRG").

Parameter name	Value
Order type	e.g. "CHRG"
Status	"N"
Data collection / control	S = None
Options / Initial status for the creation	☒

Configuration of permitted input material

Define for each output material, all [permitted input materials](#) that may be used for producing the melt of the output material. Consequently, the composition function allows for data to be restricted to the permitted materials, which simplifies the selection process.

Configuration of characteristics

Create a characteristic for each chemical element (e.g. Al, Cu, Si, Mn) that is to be considered in composition. The characteristic number and characteristic designation are only relevant to composition. The characteristics are created in the [composition recipe](#).

Configuration of composition recipe (target analysis)

Create a [composition recipe](#) (target analysis) specifying the components (see characteristics) of the material in percent (specifications: upper tolerance limit, target value, lower tolerance limit) for each input material. The target analysis defines the expected composition/make-up of an input material.

Configuration of restricted material usage

Formulas have to be defined and assigned to the recipe/material to map specific conditions (e.g. share of silicium at least 3x share of iron --> $Fe \geq Si * 3$).

This is performed within the application [material usage restrictions](#).

AIP configuration

Define the following buttons for [AIP](#): "perform charging", "confirm charging", "take sample" and "cast" in the file ctaipbut.ini.

```
[ANR-ALL-Page1]
...
6=C_CHPF,R,Perform charging
7=C_CHCF,R,Confirm charging
8=C_CHTS,R,Take sample
9=C_CHCA,R,Cast
```

The layout of the charging list is configured within the section [charge list] of the layout configuration file ctaipplay.ini:

```
[charge list]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite

MAT_VIS=C25,80,L
MATBEZ_VIS=C30,115,L
SOLL_MENGE_VIS=N12.3,78,R,target quantity
EINH=C3,40,L
RES_MENGE_VIS=N12.3,78,R,remaining quantity
VERBR_MENGE_VIS=N12.3,78,R,delivered quantity
MATPUF=C30,80,L
MATPUF_MENGE=N12.3,78,R,remaining quantity
EMAT_MENGE_VIS=N12.3,78,R,input buffer
```


3 Composition Procedure / Recomposition

Procedure/ composition process

The procedure or composition process is to be performed as follows in general.

The relevant [procedure](#) describes how to configure the required composition functions.

Generate charging order

The composition process starts, once the melting order has been received from the ERP system or by being generated from an MES work plan.

In the next step, an employee executes the function "generate charging order" (material management → composition → composition → "start composition" tab → generate charging order). This generates the charging order that is then shown in the list of charging orders including the relevant melting order. The charging order has the order status "not free". Consequently, it cannot be used/started yet at the terminal.

Start composition

The defined composition recipe including the relevant default values as well as the permitted input materials with their respective quantities is shown by the user selecting the charging order.

Composition is now started for the charging order. Consequently, the order status changes from "not free" to "reserved".

Material assignment

Once composition has started, the user can take over the bottom sump remaining in the furnace to current composition. In this case, the current bottom sump quantity will be taken over/used. The theoretical values (planned bottom sump) assumed up to now and resulting from the sequence of orders are overwritten by current information.

Then the user can assign further materials from the list of permitted materials. The user selects the required material and clicks the function "assign material" (material management --> Composition --> Composition --> tab "perform reservation" --> assign material). By double clicking the assigned material, the user can now enter the target quantity.

Subject to the added material, the list showing the selected materials is supplemented and the theoretical analysis is updated within the sample view.

Material reservation

Once material has been assigned, the materials can be reserved for the relevant charging order within the inventory (material management --> composition --> composition --> tab "perform reservation" --> reservation/perform reservation (including material)).

All or only single materials included in the list "selected materials" may be reserved for the charging order.

If the reservation is performed, the material will be reserved explicitly for this charging order and cannot be used for other charging orders.

However, it is still possible to undo or cancel reservations. Consequently, all materials are separated from the charging order and removed from the list of selected materials.

It is possible to cancel the reservation for all materials or only single materials.

In case a material has already been partially consumed, the reservation can be cancelled indeed, but the consumption will be deducted from the target quantity.

Release charging order/charging list

In case composition has been performed successfully, the charging order will be released (material management --> composition --> composition --> tab "release" --> release of charging order). The status of the charging order is "prepared". The charging order can now be used/logged on to the terminal.

The release analysis is generated by the release and saved in the system as the originally suggested charging result.

Once the charging order has been released, a "charging list" (bill of material for the charging order) can be generated by the user (material management --> composition --> composition --> tab "release" --> charging list).

Employees working in production/warehouse management are provided with the charging list to provide the relevant materials.

Procedure/ charging process

Charging is performed by an employee at the [terminal](#) pertaining to the melting furnace. To do so, the following steps are performed one after the other.

- Log charging order on
 - Log the generated charging order on
- Perform charging
 - Provision of the material in compliance with the displayed charging list from the charging order.
- The melting furnace is fed with the components and charging is completed.

- A sample is taken from the melt.
- Please also note: The result of sample taking might make re-composition necessary at MOC.
- The melt is cast. This completes the charging process.

Analysis of sampling and re-composition

The composition function enables viewing of sample results and, if necessary, to perform recomposition.

The below entries can be found in this detailed application.

Release analysis

The release analysis is the original composition result after composition has been first released and, as a result, it is the first theoretical analysis. The release analysis provides the original default values for charging and is saved/frozen as the initial status.

Theoretical analysis

The theoretical analysis first shows the current status of the used materials in relation to how the chemical make-up from the composition recipe has been achieved. Therefore, the theoretical analysis represents a default value at first.

This default value, however, is constantly recalculated, e.g. if re-composition is performed after sampling and, as a result, further materials are added to the melt.

Consequently, calculation of the current, theoretical analysis is always based on the results of actual values that are currently available in the system after sampling and not on the release analysis. This means:

- the current sample and its point in time as well as the sample weight are determined
- the actual values of the characteristics/elements of this sample are determined
- the target quantities that need to be recharged are determined (after taking the sample)
- the target material (including its make-up) is now added to the current sample's material (including its make-up) and, based on this, the composition of the theoretical analysis is calculated.

Samples (analysis based on sampling)

A sample is taken at the melting furnace. The chemical composition of the sampling is transferred to MES. The result is shown by the entry "sample" within the composition function.

If required, the user can rebblend composition and add further materials. The bottom sump cannot be taken into account with recomposition. The already used bottom sump has been consumed along with the release analysis/released composition and therefore frozen.

4 Composition Recipe

Summary

Menu	Master data → Material → Composition recipe
Transaction code	core
Function authorization	core

Utilization

The composition recipe defines the expected chemical composition of input materials and output materials.

Integration

The composition recipe is the basic prerequisite for composition. Once the charging order has been released, the recipe is used for the determination of samples. An inspection request is generated as a part of the release process. The used inspection plan is referenced in the generated inspection request.

Prerequisite

The material for which the recipe is to be created needs to be defined within the material master to be able to create composition recipes. In addition, the chemical elements to be used in the composition recipe also need to be defined.

Selection criteria

The paragraph that follows shows some of the available selection criteria. Self-explanatory filter options are not listed.

Area

"Composition recipe" is set here by default.

Active

By checking this checkbox, the list of composition recipes can be restricted to active recipes. If this checkbox is not checked, the list only shows composition recipes in the status "in process" and "released". The third state of this checkbox (grayed out) shows all composition recipes. This is the initial state.

Recipe number

Filters the recipe numbers of composition recipes.

Recipe version

Filters the recipe version of composition recipes.

Field descriptions

Area, recipe number, recipe version

The "area", "recipe number" and "recipe version" uniquely identify all existing composition recipes. The area is set to "composition recipe". The recipe number and recipe version may be entered using alphanumeric characters. All these fields are mandatory fields.

By assigning a structured recipe number, it is possible to provide specific information. This information might be useful later during sorting. If an existing recipe version is to be modified, yet it cannot just be changed because it has already been used for the generation of inspection orders, it is recommended to copy the original composition recipe and to modify the recipe version (e.g. incrementing it by 1). The recipe number should be kept as far as possible.

Material number

Shows the material number. If it is known it can be entered directly. Otherwise, the material dialog can be opened and the provided filter and sort criteria may be used to identify and take over the required material. Once a material has been chosen from the master data record, the material designation, customer article number and drawing issue number are taken over and displayed in the relevant fields.

Released/active

Shows whether the inspection recipe is "released" and/or "active". If the recipe is released or active the corresponding checkboxes are checked. A recipe is released and enabled, i.e. its status is changed, only by using the corresponding toolbar functions. A recipe has to be released before it can be activated.



Inspection orders are generated in the system to perform composition. Please note that the automatic generation of inspection orders only considers released composition recipes.

Released by / on

Shows the HYDRA user who has released the recipe. The release date is displayed additionally.

Valid from / until

If required, a validity period may be entered here, instead of the "unrestricted" activation (using the toolbar). This period is then taken into account when the inspection order is generated. Yet activation for a certain period means that the user has no clear overview of currently valid inspection plans, and it is therefore recommended to use the "global/unrestricted" activation option using the toolbar. If activated by toolbar functions, the system carefully monitors whether an active inspection plan already exists for the specified article and it also includes the same drawing issue number, customers and suppliers. If this is indeed the case, the previously active inspection plan will automatically be disabled.

Editing functions

The key fields "area", "recipe number" and "recipe version" cannot be changed in the editing mode.

Toolbar



Copy

A corresponding dialog opens for copying of a composition recipe.

The target area type and target area may be entered here. Normally, the user should choose an area that is identical to that of the source inspection plan. Then the new recipe number and recipe version need to be entered. In case a new version is generated from an existing recipe, normally the same recipe number is used and only the recipe version is changed.



Activate

Function authorization: core.activate

Makes the composition recipe status "active".



Deactivate

Function authorization: core.deactiv

Puts the composition recipe that is in the "active" status back to the "released" status.



Release

Function authorization: core.release

Puts a composition recipe that is in the "in process" status to the "released" status.



In process

Function authorization: core.unreal

Puts a composition recipe that is in the "active" or "released" status to the "in process" status.

Detail application "print form"

Function authorization	core.print
------------------------	------------

The print dialog opens a list of available reports. These are Word forms. The potential content of these forms is determined by the Web services that are available in the respective context. The form entries, i.e. the contents of the list of forms of the corresponding print dialog, are defined within the master data of quality management. The basis for new forms and the corresponding form properties are defined there as well. A corresponding license is required to be able to change the forms as regards content and design.

Detail application "characteristics"

The detail application for characteristics is nearly identical to the master data of characteristics application. For this reason, reference is made here only to modifications or additional features.

The relevant characteristics are assigned to the previously defined composition recipe on the level of characteristics. Characteristics are assigned by creating a new data record and by opening the characteristic catalog and accepting the characteristic selected there. All master data entries are copied into the characteristic, once the characteristic has been taken over. Each (copied) information can be changed and/or amended afterwards. Characteristic designations are often supplemented to define the characteristic in more detail.

It is also possible to create a characteristic that is not included in the characteristics catalog. However, this is recommended only in exceptional cases, since all analyses (e.g. failure mode analysis) are based on characteristics included in the catalog. It is therefore recommendable to maintain the characteristics catalog properly.

Different properties and settings can still be defined, before specific characteristic data is supplemented.

Field descriptions

Position

The position determines the order of subsequent inspections. The input must be unique. Ideally, the position number should be incremented in steps of ten when new data records are created. Consequently, a new characteristic may still be inserted between two existing characteristics at a later point in time.

Characteristics number

Number of the characteristic selected from master data.

Characteristic designation

Designation of the selected characteristic.

Detail application "inspection plan documents"

As many documents as required may be assigned to each composition recipe, provided that the "inspection plan documents" tab has been enabled in the master detail grid. By enabling these tabs, the toolbar provides corresponding buttons to edit documents.

All formats registered by Windows are available, when documents are assigned. Consequently, simple documents (e.g. written in Word), drawings of any format and videos may be assigned. However, the corresponding programs that are able to display the required formats have to be installed. In this context, the documents are opened by the program that has been linked in Windows.

The file types "FILE", "URL" and "Text" are provided. The file name including path may be entered manually with the "file" type. The "URL" file type allows access to the Internet or Intranet. The third file type "Text" allows for text to be entered directly.

A designation may be assigned to each defined document. Moreover, it may also be determined in which order the documents are to be listed. The "position" field is used for this purpose (numeric input). The specifications made within this list must be unique. In addition, the checkbox "display during inspection" specifies whether or not the document may be shown during the inspection process.

5 Material Usage Restrictions

Summary

Menu	Master data → Material → Material usage restrictions
Transaction code	rest
Function authorization	rest

Utilization

Material usage restrictions (composition restrictions) include formulas that are used for checking the composition (e.g. the relation of materials) for the production of a material (alloy).

Integration

The composition for the charge make-up is calculated or analyzed based on the material usage restrictions.

Selection criteria

Material

The material for which material usage restrictions are to be shown.

Formula

The formula specifying a restriction

Comment

Comment to explain the usage restriction.

Description

Brief description of the usage restriction.

Field descriptions

Material

The material to which the usage restriction refers.

Formula

The formula characterizing the restriction. The formula expression is shown in addition to the formula key. Formulas of the type = 7 "formulas for composition restrictions" or type = 8 "formulas for composition scrap rate" are used for composition.

Comment

The comment assigned to the formula.

Usage type

Only option "C" for composition is supported at the moment.

Description

Brief description of the usage restriction.

Comment

Comment to explain the usage restriction

Editor

The user who edited the data record at last

Modified on

The time when this data record was edited at last

How to use formulas

The below-mentioned variables may be used in formulas

"Material"

All components/elements (defined as characteristics in composition recipes) can be used with their names in formulas

SCRAPMATERIAL

Total target quantity of scrap material in the selected materials

RAWMATERIAL

Total target quantity of raw material ("non" scrap material) in the selected materials

SUMP

Target quantity of the current bottom sump

MAXCAPACITY

The maximum capacity of the melting aggregate(s)

Examples

This section shows some examples for formulas

Share of iron max.1/2 share of copper: $Fe \leq (Cu / 2)$

The variables (here: Fe und Cu) are assigned based on the components/elements of the materials to be used

Scrap rate max.40%: $SCRAPMATERIAL / (SCRAPMATERIAL + RAWMATERIAL) < 0.4$

The target quantity for scrap material (SCRAPMATERIAL) relating to the total quantity (scrap material plus raw material) must not reach 40%

Max. capacity: $SCRAPMATERIAL+RAWMATERIAL+SUMP<MAXCAPACITY$

The target quantity for scrap material (SCRAPMATERIAL) plus raw material (RAWMATERIAL) plus current sump (SUMP) must be less than the maximum capacity of the melting aggregate(s) (MAXCAPACITY).

6 Permitted Input Materials

Summary

Menu	Master data → Material → Permitted input materials
Transaction code	pema
Function authorization	pema

Utilization

This application defines the input materials permitted for composition.

Integration

Only input materials defined as "permitted input materials" may be used in composition for manufacturing a material (an alloy). If an input material has been identified as additional material, it may also be used for re-composition.

Composition is supported by the defined, permitted input materials.

Prerequisite

Selection criteria

The following selection criteria are available in the application:

Material

Material for which a permitted input material has been defined.

Input material

Input material of composition.

Additional material

Selected data records that have been identified as additional material.

Comment

Comment to explain the relation

Description

A brief description of the relation

User fields

The defined user fields of the object type MATZEMAT

Field descriptions

These parameters are shown:

Material

Material for which a permitted input material has been defined.

Input material

Permitted input material for the (end) material

Additional material

The input material has been defined as additional material for the (end) material and may, for example, be used for re-composition.

Comment

Comment to explain the relation

Description

A brief description of the relation

Editor

The user who edited the data record at last

Modified on

The time when this data record was edited at last

User fields tab

User fields offer the possibility to store further customer-specific information to MES besides the fields available in MOC standard. The tab provides eight sub tabs each of which providing eight user fields. The so called user field key determines which user fields are involved and which meaning they have.

Object type

Default "MATZEMAT"

User field key

Default "SYSTEM"

User fields

The following user fields are available after customizing the system:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16
Text field, length 10	6
Text field, length 20	14
Text field, length 40	2

A maximum of 8 fields are shown for each page.

User field keys are not defined by default in the system. The system has to be customized accordingly to be able to support this kind of user fields.

7 Material Master

Overview

Menu	Master data → Material → Material master
Transaction code	matc
Function authorization	matc

Available user fields		
Where?	Object type/user field key	Source (type)
Detail view	ARTIKEL/SYSTEM	Material (MF-D)
How to configure user fields? Which user field types are available?		

Purpose

You use this function to create a material master in the system for the materials used.

Integration

The material master has been designed to edit materials. This refers to especially defined master data. The material data defined in the master data are used by the composition function or e-kanban systems.

Requirements

When you create the master data for a material, the material type and the material buffer must already exist in the system.

Selection criteria

The application provides the following selection criteria:

Tab *General*

Material number

Material number

Drawing issue number

Drawing issue number of the material, also referred to as index

Designation (name)

Material name

Inactive

Inactive, active materials. The checkbox is not enabled by default.

Tab *Composition***Scrap material**

Selects materials identified as scrap material.

End product

Selects materials identified as end products.

Input material

Selects materials identified as input material. The checkbox is enabled by default.

Tab *Kanban*

To show these fields, you require the user authorization "kov".

Kanban material

Selects the materials used in the kanban process.

Tab *User fields*

You can use this tab to perform a selection according to the defined user fields for object type "ARTIKEL" and the user field key "SYSTEM".

Field descriptions**Material number**

Material number

Drawing issue number

Drawing issue number of the material, also referred to as index

Designation (name)

Material name

Material type

Material type of the material

Specific weight

The specific weight of the material in the unit g/mm³

Inactive

Inactive, active materials. The checkbox is not enabled by default.

Input material

The material is an input material of the composition. This option has to be enabled for materials used in composition.

Scrap material

Identifies a material as scrap material.

Material buffer

Material buffer of the material

Fragmented size

The material piece size in kg.

Price

The price of the material in €/kg

End product

Identifies a material as end product of the composition.

Tab Kanban

To show these fields, you require the user authorization "kov".

Kanban material

Identifies a material as kanban material.

Tab MSL (only with HYDRA for Electronics)

The Moisture Sensitivity Level (MSL) specifies the moisture sensitivity of semiconductor components during packing, storage and assembly.

MSL material

This option specifies if the MSL level is relevant to material of this material type.

MSL term

Time period in hours in which the material must be used after opening.

Level/threshold

Specifies the level of the MSL material.

Comment

You can enter a comment in this field.

Tab *User fields*

You can use user fields to store additional customer-specific information in the MES. The *user fields* tab includes eight sub-index tabs, which each has eight additional user fields. The so-called user field key specifies the available user fields and their meaning.

Object type

Default "ARTIKEL"

User field key

Default "SYSTEM"

User fields

The following user fields are available after system customization (maximum):

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16
Text field, length 10	6
Text field, length 20	14
Text field, length 40	2

Each page shows a maximum of 8 fields.

By default, no user field keys are defined. You must activate in the system via customization that this kind of user fields is supported.

Toolbar



Composition recipe

This function calls the relevant composition recipe for the selected material.

8 Work Plan - Edit Order Components

Summary

Menu	Order management → Routing management → Work plan – Edit components of the order
Transaction code	edworcmp
Function authorization	edworcmp

The "work plan - edit components of the order" application allows for the material components, which are required to produce the article in the current manufacturing level (current operation), to be displayed and edited.

Normally, these components are transferred to HYDRA using an interface from the higher-level ERP system, as these components are already defined in the ERP work plan.

Selection criteria

The following selection criteria are available in the application:

Order

The components assigned to a work plan order may be selected by entering an order.

Field descriptions

The fields pertaining to components are described [here](#)

Editing functions

Please use the available buttons to create new or edit existing work plan components. A copy function for components is not planned.



Please note that the BOM item must be **unique** within the operation if HYDRA-MPL is in use!

Toolbar



Edit operations

Function authorization: edwop

Opens the application [Work plan – edit operations](#).

**Edit orders**

Function authorization: edwor

Opens the application [Work plan – edit orders](#).

9 Work Plan - Edit Production Resources & Tools

1.1 Summary

Menu	Order management --> Routing management --> Work plan - Edit production resources & tools
Transaction code	edwres
Function authorization	edwres

The "production resources & tools" application allows for the resources, which are required to produce the article in the current manufacturing level (current operation), to be displayed and edited.

Production resources and tools may be, for example, tools, documents, NC programs, etc.

Selection criteria

The application provides the following selection criteria:

MES work plan number

The production resources and tools assigned to a work plan operation may be selected by entering the MES work plan number. The MES work plan number is the combined work plan/operation number.

Enter the whole MES work plan number if you would like to view the production resources & tools assigned to a specific operation.

If you would like to view the production resources & tools of all operations of a work plan only enter the work plan number, followed by "**".

Field Descriptions

The fields of a production tool and resource are described [here](#).

Editing functions

Please use the available buttons to create or edit production resources & tools of work plans. A copy function for production resources & tools is not planned.



If the tool and resource management module (HYDRA-WRM) is in use, the first production resource and tool that is *not* of the resource type "DNC" or "MAT" is taken over into the "tool" field of the [operation](#). In addition, the "tool" field is checked whether it already includes a value, when inserting a production resource and tool that is not of the "DNC" or "MAT" resource type. If this is not the case, this component is taken over. For this reason, it is recommended to insert

the "main production resource & tool" at first in the list of production resources and tools.



Please note with regard to documents: If a new document is assigned to an operation a file is only uploaded automatically, in case a file has been selected using the file selection dialog. The file selection dialog can be opened by the button next to the "file name" field.

In this case, the path of the file that is loaded onto the server is displayed below the input field for the file name. The upload is performed automatically while saving.

No file can be uploaded if the file name is entered manually.

The corresponding data record is created anyway even if an error occurs during the upload.

Toolbar



Edit operations

Function authorization: edwop

Opens the application [work plan - edit operations](#).



Edit orders

Function authorization: edwor

Opens the application [work plan - edit orders](#).

- →→→→

10 Edit Order Components

Summary

Menu	Order management → Order management → Edit components of the order
Transaction code	edorcomp
Function authorization	edorcomp

Utilization

Materials required for manufacturing an article/item are assigned to an operation or order as "(material) components". This function allows for material components of an order to be displayed or edited.

Normally, these components are transferred to HYDRA using an interface from the higher-level ERP system, as these components are already defined in the ERP work plan.

Prerequisite

The relevant order has to be created.

Selection criteria

The following selection criteria are available in the application:

Order

The components assigned to an order may be selected by entering an order.

Field descriptions

Order

The order number to which the production resources and tools are to be assigned can be entered in this field.

Material

The material number of the material component can be entered here.

Designation

The material designation can be entered in this field.

Comment 1 / comment 2

These are comment fields.

BOM item

In a bill of material the components of a product are referred to as items. The number entered here arranges the BOM items within the BOM. Consequently, identical material numbers may occur in the component list but data will still be assigned to the correct position of the component.

When it comes to reel-based manufacturing, it refers to the position of the component within the layer structure.

Please note that the BOM item is an integral part of the identification key.

BOM level

A component can also have several levels. Enter the BOM level here, provided that it is available and known.

Please note: Postings can only be made for materials of the BOM level 0. If a BOM level > 0 is entered, the component type (see next field) will generally be set to "I" (information component).

Component type

Component type. Possible values are:

M	Material component. The component type is to be indicated here. The other types might also be relevant for MPL or its reel-based solution MPL-RF.
I	Info component. Information components can be shown in the component list without having to be posted.
T	Carrier material (MPL-RF). A maximum of one input batch may be logged on to the machine as carrier material (T) or added material (Z).
Z	Added material as alternative for the carrier
A	Scrap/waste material

Material type

Material type of the material component. The material type controls processing specific to the material in HYDRA.

Unless otherwise stated in the project, the material type SYSTEM is to be assigned.

The material type must exist in HYDRA. If no material type is entered, HYDRA tries to determine the material component (prerequisite: the assignments of material to the material type are up-to-date). If no material type is found HYDRA assigns the material type SYSTEM.

Please note: A separate material type (e.g. INFO) should be defined and assigned for information components (material type "I").

Consumption type

The following input options are planned for material components. The definition of individual options and their usage depends, among other things, on which HYDRA modules are in use.

K = None

This option specifies that no consumption is collected for this material component. In this case, the material component is only displayed.

"Info components" (see above: material type) have to be set to this option.

D = Discrete

With this component, the consumption is determined in a retrograde manner at the Windows terminal, i.e. based on the last produced quantity and suggested in the posting dialog. The consumption is posted for the component and a material movement (goods issue from production) is generated that can be uploaded to the higher-level ERP system (however, this requires the relevant interface for uploading goods movements).

This type of material consumption recording needs to be configured especially while HYDRA is customized.

L = with batch reference (relevant if MPL/TRT is in use)

This option results in the material component to be logged on and off as HYDRA batch. Calculating the consumption for this material component (retrograde, with logging the input batch off) depends on the configuration of the material type to which the material component is assigned.

Required quantity/unit

Planned total quantity of the component within the operation.

It is calculated automatically from the target quantity of the operation (primary quantity unit) multiplied by the input quantity of the component.

The required quantity is only shown in the table as well as in the detail panel.

Input quantity

Planned input quantity of the component for each unit of the operation's primary quantity.

MPL, consumption type D: Planned input quantity of the component for each unit of the operation's primary quantity

Unit

Quantity unit for the input quantity

Input quantity in % / upper tolerance limit / lower tolerance limit

Reserved. No processing. Should be set to 0.

Replaceable

If this flag is set, another than the planned material may be used for the component. However, in this case, only material of the same material type can be used.

Requirement to change output batch

The input batch change for a batch of this material also requires an output batch change. Please note in this context:

☒, if type = T or Z

☐, if type = I or A

☒☐, if type = M

Otherwise: N

Superior component: BOM item/BOM level

Reserved. No processing.

Toolbar



Edit operations

Starts the application [edit operations](#).



Edit orders

Starts the application [edit orders](#).



Order information

Starts the application [order information](#).

11 Edit Production Resources and Tools

Overview

HYDRA menu	Order management → Order management → Edit production resources and tools
FEDRA menu	Detailed Scheduling → Order management → Edit production resources and tools
Transaction code	edres
Function authorization	edres

Purpose

Resources can be defined for operations in the list of production resources and tools.



Further information on how to define workforce requirements via production resources and tools can be found in the document entitled [Definition of Workforce Requirement.pdf](#)

Requirement

The corresponding operation must already be defined.

Selection criteria

The application provides the following selection criteria:

MES order number

Combined order/ operation number.

Please note that the components are assigned by specific operations. This is why the entire key must be entered. By entering the order number followed by *, the system will list all components for an entire order.

Field descriptions

Order/ operation

Enter the order/ operation number for the operation that is to be assigned to the production resource or tool here.

Resource type

Resource type of the production resource or tool that is to be assigned to the operation. The resource type must be known in the system. Predefined resource types must be chosen from the selection menu. Additional resource types can be defined when customizing HYDRA. For documents, the resource type to be entered here must be DOC.

Resource

Enter the resource number (material number) of the production resource or tool.

Designation

Here, you can enter a name for the production resource.

Comment 1/ Comment 2

These are comment fields.

Required quantity/ unit

Resource quantity required to carry out the operation. When planning the operation in the shop floor scheduling, this number of resources is entered in terms of capacities. The quantity unit is only used as a comment.

Please note: In the shop floor scheduling, the quantity 0 is interpreted implicitly as quantity 1.

Path

When identifying a document as a production resource, the logical reference to the path is to be defined in the path configuration (menu: File > System administration > Paths). No path must be stored for DNC resources; it is determined based on the path stored for the resource type. The field should be left empty for all other production resources (only applies when using HYDRA).

File

When identifying a document as a production resource, the file name (including file extension) is to be entered here.

No file name must be stored for DNC resources; it is determined based on the file name defined for the resource. The field should be left empty for all other production resources (only applies when using HYDRA).

Modified by/ date/ time

Editor as well as the date and time the last change was made.

Please note with regard to documents: If a new document is assigned to an operation a file is only uploaded automatically, in case a file has been selected using the file selection dialog. The file selection dialog can be opened by the button next to the “file name” field.



In this case, the path of the file that is loaded onto the server is displayed below the input field for the file name. The upload is performed automatically while saving.

No file can be uploaded if the file name is entered manually.

The corresponding data record is created anyway even if an error occurs during the upload.

Toolbar



Edit operations

Calls the application [Edit operations](#).



Edit orders

Calls the application [Edit orders](#).



Order information

Calls the application [Order information](#).

12 Characteristic Master Data

Overview

Menu	Master data → Quality management → Characteristics
Transaction code	chrq
Function authorization	chrq

Available user fields		
Where?	Object type/user field key	Source (type)
Table and detail view	CMM/SYSTEM	QM
How to configure user fields?		Which user field types are available?

Purpose

The catalog of characteristics has been designed to define characteristics and, as a result, to predefine characteristic data of inspection plans. For this reason, it aims at people involved in inspection planning.

The catalog of characteristics is one of the most important basic catalogs. You cannot set up any inspection plan without this catalog. As this catalogue is used to predefine characteristics for inspection plans, it includes extensive input options. Basically, the catalog of characteristics should only include such data, which will not have to be modified when the characteristics are assigned to the inspection plan later on. For example, the definition of limit values is usually not reasonable as these values are only known when an inspection plan is set up. Only when you assign data to an inspection plan, a relation between data and article is established. Note this and you will know what kind of information you should predefine. For example, it must be carefully considered whether the characteristic "outer diameter" is only created once and detailed information is stored in the inspection planning later on or whether several "outer diameter characteristics" are created, e. g. with specification of limit values. Usually, it is an advantage to store a restricted number of general characteristics. The required evaluations/reports also play a role in this context. If a new "outer diameter characteristic" is created for almost every tolerance change, this characteristic is "valid" for one article only. In a subsequent failure analysis, a comprehensive evaluation is not possible in this case!

It is important that each detail defined here can be modified in the inspection planning later on or that details, which have not been stated, can still be added.

The configurations made in the characteristics' master data are not final. The characteristics' master data is used as a template for later inspection planning. You can complete and modify all settings of the characteristics' master data during inspection planning.

Integration

The catalog of characteristics is a global catalog that is used in many QM applications. Please find below some possible fields of application that refer to the catalog of characteristics.

- Inspection planning for production, goods receipt, goods issue, initial samples and calibration
- Inspection requirements for production, goods receipt, goods issue, initial samples and calibration
- Failure analysis in complaint management
- Several reports/evaluations

Requirements

There are no special requirements.

Selection criteria

The application provides the following selection criteria:

- Characteristic no.:
Number of the characteristic
- Characteristic designation:
Designation of the characteristic – Note: You may use wildcards "**"
- Characteristic type:
Inspection type: attributive, inspection chart, variable

Tab Details

- Gage
Select a gage
- Gage designation:
Select a gage designation

Tab User fields

- If user fields are created, they may be selected

If several selection criteria are used, overlapping results are displayed in the characteristics' master data.

In addition, the column filter allows for the content of each individual column to be filtered.

Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

Tab Characteristics

Characteristic no.

Unique number of the characteristic

Characteristic designation/name

Designation of the characteristic

Input type

Automatic or manual data collection. This field controls the release of HYDRA-PDV fields (in case of automatic collection). If the automatic collection function is selected, the characteristic type is restricted to the "variable" option.

Characteristic type

This option specifies whether the collection of measured values (variable) or the identification of the number of detected failures (attributive) is used for the inspection. If you select the attributive inspection, use the input type to define whether the collection should be based on a catalog or whether the standard collection is performed. Further characteristic types are the inspection chart and the information characteristic. If you select the inspection chart, you can enable the input type *visual defects recording*. The information characteristic is only used to display a document during the inspection process. Subject to the input type, the lower area of the dialog provides the respective sampling schemes.

Visual recording: The characteristic document (not the *inspection requirement document*) is displayed with the position 1. This must be type FILE. The system supports these formats: JPEG, JPG, PNG. To divide a graphic in different areas, you must define the grid for the x-axis and the y-axis (e.g. A,B,C,D,E)

Inspection result base

This setting defines whether all samples or only the sample recorded last is used to identify the inspection result (pass/fail).

Mandatory inspection

If this option is activated, you must enter at least one measured value for this characteristic, before you can complete an inspection order including this characteristic.

Formula:

See chapter Calculation of formulas.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.formula".

Tab Details

Group Gage

Gage

Defines whether a gage or gage group is to be assigned to the characteristic:

Assignment of the gage (or gage group) to be used.

You can also use resources of resource type "PRM" of the resource management. To display this field in the inspection plan characteristics, you require the authorization "iriscp.gage".

Gage designation (name of test equipment)

Shows the name of the gage

Group Properties

Certificate printing

The selected option defines whether this characteristic is to be printed (display selection or print always) or not (print never) when certificates are printed at a later stage (e.g. acceptance, inspection certificate). If you select the option "display selection", a list of the characteristics with this printing option set is displayed prior to printing. In the list, these characteristics are preselected for the print of a certificate. However, this selection may be removed. Finally, all selected characteristics and the characteristics with the "print always" option are included in the certificate print. Characteristics with the "print always" option do not appear in a selection list, as they are printed in any case. Please note that this option only affects certificate forms.

Failure weighting

If the inspection result for the characteristic is "fail", you can classify the result here for information purposes.

Group Inspect

Analyseauswahlkatalog

Here, you can select an analysis selection catalog. The catalog restricts the selection of possible failures you can enter (failure types, failure location, etc.). (All available failures may still be entered, if you directly enter their number).

Designation of analysis selection

Shows the designations of analysis selection catalogs

Tab Specifications

Once the "specifications" tab has been selected, the sample scheme and constructional measures may be entered. In this context, it has to be considered that (as already mentioned) the definition of tolerance limits in the master data of characteristics is only reasonable if certain conditions are met. The same applies to the definition or calculation of action and warning limits. This section explains the possibilities in detail.

Group Sampling scheme

Sampling scheme

The following sampling schemes are available:

- 100% inspection
- k value inspection
- lot inspection
- n-c inspection
- SPC inspection

The sampling scheme defines the inspection procedure. In case of an n-c inspection and parameters 5-0, 5 pieces are checked and 0 failures may be detected.

Find a more detailed description in section Sampling **schemes**.

Sample size/expected sample size

Specification of the sample size (number of samples) or the expected sample size depending on the sampling scheme, see section Sampling schemes.

Acceptance quantity

Acceptance quantity for the n-c inspection, please also see section Sampling schemes.

Interval type

Input for SPC or n-c inspections: time, pieces, once, none. See chapter Sampling schemes .

Interval value

Specifies the interval subject to the interval unit.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Interval unit

For n-c or SPC inspections, e.g. minutes, hours.

With output batch change

If the output batch changes, an inspection becomes due.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Note:

The option *With output batch change* only triggers the generation of an inspection point, if the respective change of the output batch is included in the dialog "Change of batches" (dialog ID: CA_WL). For example, reel cutting dialogs do not generate inspection points.

With machine status change

If the machine status changes, an inspection becomes due.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Source status

Here, you can specify source statuses (specific non-productive machine statuses) – separated by commas. If the machine then changes from a specified source status into a productive machine status, an inspection becomes due.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

As of SP8, the following configurations are available in addition.

- The field is completely empty: For this characteristic, a machine status change always generates an inspection point, if the machine changes from a non-productive into a productive status.
- "x-y", comma-separated: If the machine changes from source status x to target status y (may be non-productive), an inspection point is generated for this characteristic.
- "x-": If the machine changes from source status x to an arbitrary target status (may be non-productive), an inspection point is generated.
- "-y": If the machine changes from an arbitrary source status to target status "y" (may be non-productive), an inspection point is generated.

With change of shifts

An inspection becomes due on changing shifts.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Inspection due date of last off inspection

For details on the configuration of a last off inspection, refer to the section "[Last off inspection](#)".

Group *Constructional measures*

Unit

Pieces, meter, kg, etc. Unit of the characteristic. Allocate the units by using the unit catalog.

Decimal places

Number of decimal places. Leading zeros before the comma are not displayed in the specification fields. By default, the number of decimal places defined in the system settings is pre-assigned.

Size (measure type)

Plausibility and tolerance limits can be entered as absolute, relative or percentage values. Please note that relative or percentage lower limits (lower tolerance limit, lower process limits) must be specified with a negative algebraic sign.

Standard

Calculation of tolerances based on specific standards (e.g. ISO metric fits). Subject to the selected standard, further information is requested (e.g. engineering fit). The system automatically calculates the tolerance limits on the basis of these specifications.

Fit

Calculation of tolerance limits on the basis of a specific standard and engineering fit. The selected fit depends on the selected standard.

Upper PL

Specifies the upper plausibility limit

Upper TL

Specifies the upper tolerance limit (upper specification limit)

Target value

Specifies the target value

Lower TL

Specifies the lower tolerance limit (lower specification limit)

Lower PL

Specifies the lower plausibility limit

Generate failure (UTL)/(LTL)

If measured values are recorded and the checkbox *Generate failure* is enabled, a violation of the limit value automatically results (in the background) in the failure type "limit value violation" (AUTO:TG> or AUTO:TG<). This option is not available for attributive characteristics, as the specification is only used for information purposes in this case.

User fields tab

If you have defined user fields for characteristics, they are displayed and may be edited here.

Tab Chart 1/Chart 2

In tab chart1/chart2, you can define the control charts to be used. These control charts are later available in the integrated measurement recording and in the measurement recording for terminals (SPCM). You can define a total of two different control charts. Here, you can store for each control chart the action limits, warning limits and the mean value of variable characteristics. There are two different possibilities to define these limit values. You can enter the limit values manually or the limit values are calculated using the specified default values included in tab *Default values chart1/2*. For further information on control charts, refer to sections 12.2 Control charts for variable characteristics and 12.3 Control charts for attributive characteristics.

Chart 1 / Chart 2

Specifies the control chart displayed in the measurement recording dialog on the terminal. You can define action limits on the basis of the control chart type.

Upper AL

Specifies the upper action limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

Upper WL

Specifies the upper warning limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

MV (Mean value)

Specifies a mean value, e.g. as basis for the automatic calculation of limits by the system.

Lower WL

Specifies the lower warning limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

Lower AL

Specifies the lower action limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

Generate trend error

The option "generate trend error" has to be activated to be able to generate an automatic error if a trend exists (e.g. seven values in a row are descending or ascending, the number of values is defined while the system is customized). To identify a trend, the samples of an inspection step characteristic are checked, sorted by their sample number – regardless of the machine where the data has been recorded.

Generate error (UWL) / (LWL)

Enable the checkboxes *Generate error (UWL) / (LWL)* to generate automatically (in the background) the failure type "Limit value violation" (AUTO:WG> or AUTO:WG<), if a limit value is violated during the recording of measured values. Here, the violation of the limit value is identified using the stored control chart. In case an xq chart is stored, the automatic error is only generated if the respective xq value of the control chart, and not the single value, exceeds the warning limits.

Generate error (UAL) / (LAL)

Enable the checkboxes *Generate error (UAL) / (LAL)* to generate automatically (in the background) the failure type "Limit value violation" (AUTO:EG> or AUTO:EG<), if a limit value is violated during the recording of measured values. Here, the violation of the limit value is identified using the stored control chart. In case an xq chart is stored, the automatic error is only generated if the respective xq value of the control chart, and not the single value, exceeds the action limits.

Tab **Default values chart 1 / Default values chart 2**

For further information on control charts, refer to sections Control charts for variable characteristics and Control charts for attributive characteristics.

Group Default for calculating limit values**Calculation type**

Default values to calculate limit values: Cpk, Sigma, sq/an, Rq/dn, relative deviation from xq, deviation from xq in percent

Cpk

Default value of cpk

Sigma

Default value or calculated sigma value

Rq/sq (RQuer/sQuer)

Default value for Rq/sq (RQuer/sQuer)

Group Non-action probability**Action limits (non-action probability)**

Specifies the action probability (only visible with the calculation type: cpk, Sigma, rQuer/sQuer)

Warning limits (non-action probability)

Specifies the action probability (only visible with the calculation type: cpk, Sigma, rQuer/sQuer)

Group Deviation from xq specification**rel. AL**

Direct entry of the action limits (only visible with calculation types relative/percentage deviation).

rel. WL

Direct entry of the warning limits (only visible with calculation types relative/percentage deviation).

Group Confidence interval**Confidence interval**

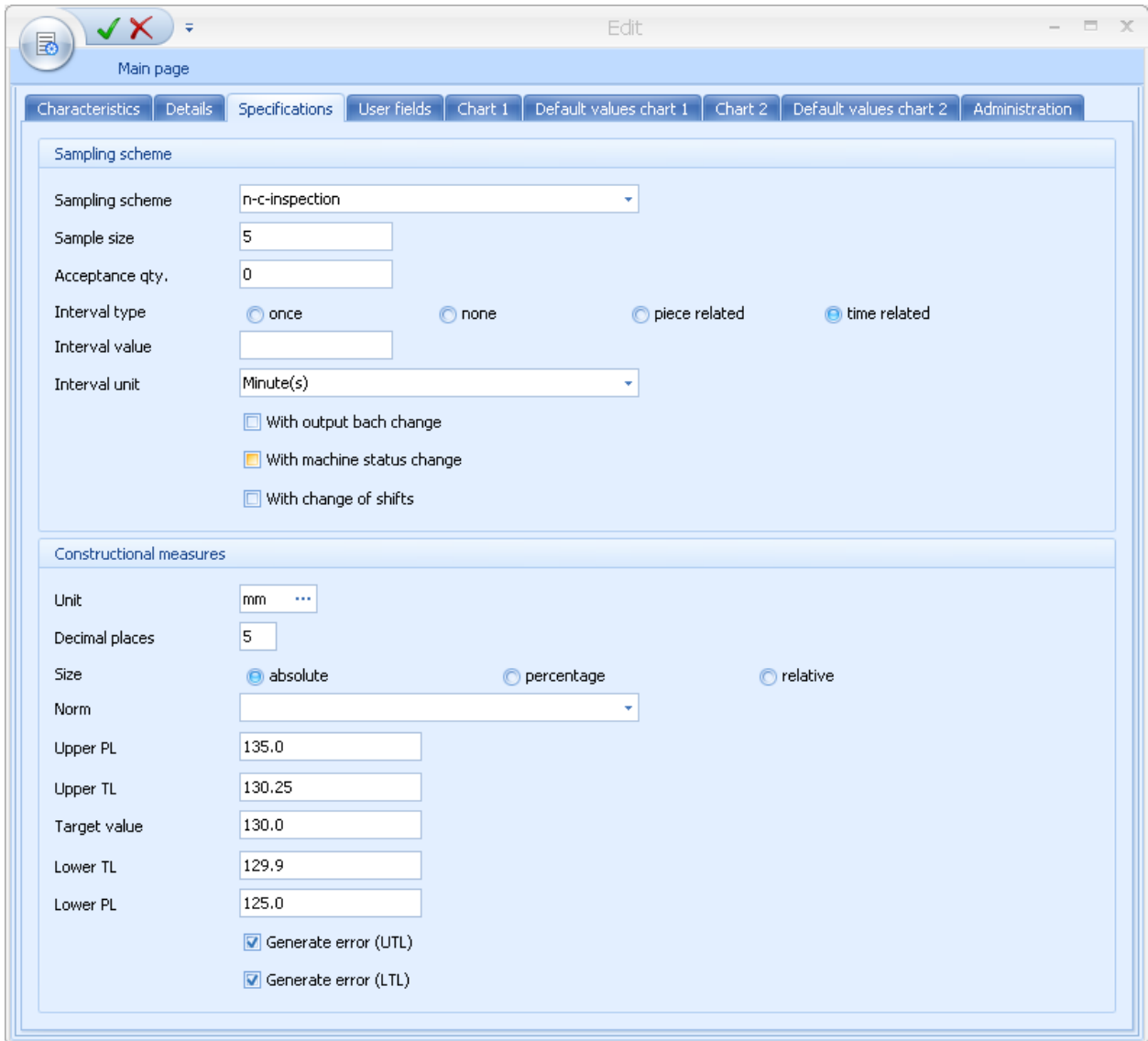
One-sided or two-sided. You can select one-sided or two-sided for the control charts R and s.

Group xq**XQ**

Target value, mid-tolerance, mean value of xq chart, input (only visible and can only be selected with an xq control chart)

Editing functions

The below screenshot shows an example of an editing dialog. Design and alignment of fields may deviate.



The screenshot shows a software window titled 'Edit' with a standard Windows-style title bar (minimize, maximize, close buttons). The window has a tabbed interface with the following tabs: 'Main page', 'Characteristics', 'Details', 'Specifications', 'User fields', 'Chart 1', 'Default values chart 1', 'Chart 2', 'Default values chart 2', and 'Administration'. The 'Specifications' tab is currently selected.

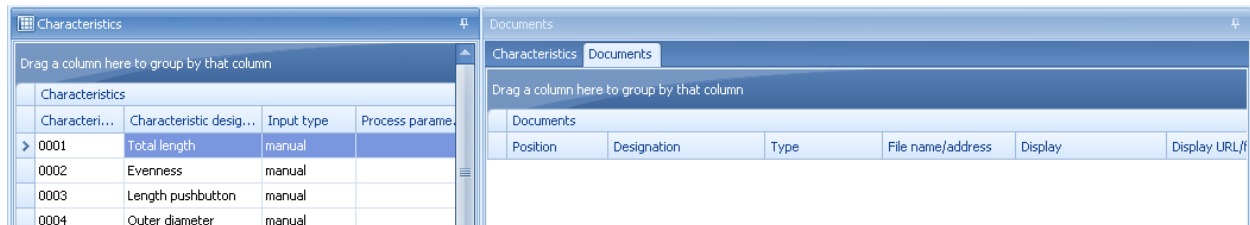
The 'Specifications' tab is divided into two main sections:

- Sampling scheme:**
 - 'Sampling scheme': A dropdown menu showing 'n-c-inspection'.
 - 'Sample size': A text input field containing '5'.
 - 'Acceptance qty.': A text input field containing '0'.
 - 'Interval type': Four radio buttons: 'once' (selected), 'none', 'piece related', and 'time related'.
 - 'Interval value': A text input field.
 - 'Interval unit': A dropdown menu showing 'Minute(s)'.
 - Three checkboxes: 'With output batch change' (unchecked), 'With machine status change' (checked), and 'With change of shifts' (unchecked).
- Constructional measures:**
 - 'Unit': A text input field showing 'mm' with a small menu icon to its right.
 - 'Decimal places': A text input field containing '5'.
 - 'Size': Three radio buttons: 'absolute' (selected), 'percentage', and 'relative'.
 - 'Norm': A dropdown menu.
 - 'Upper PL': A text input field containing '135.0'.
 - 'Upper TL': A text input field containing '130.25'.
 - 'Target value': A text input field containing '130.0'.
 - 'Lower TL': A text input field containing '129.9'.
 - 'Lower PL': A text input field containing '125.0'.
 - Two checkboxes: 'Generate error (UTL)' (checked) and 'Generate error (LTL)' (checked).

Toolbar

There are no other special function buttons in addition to the standard functions/features.

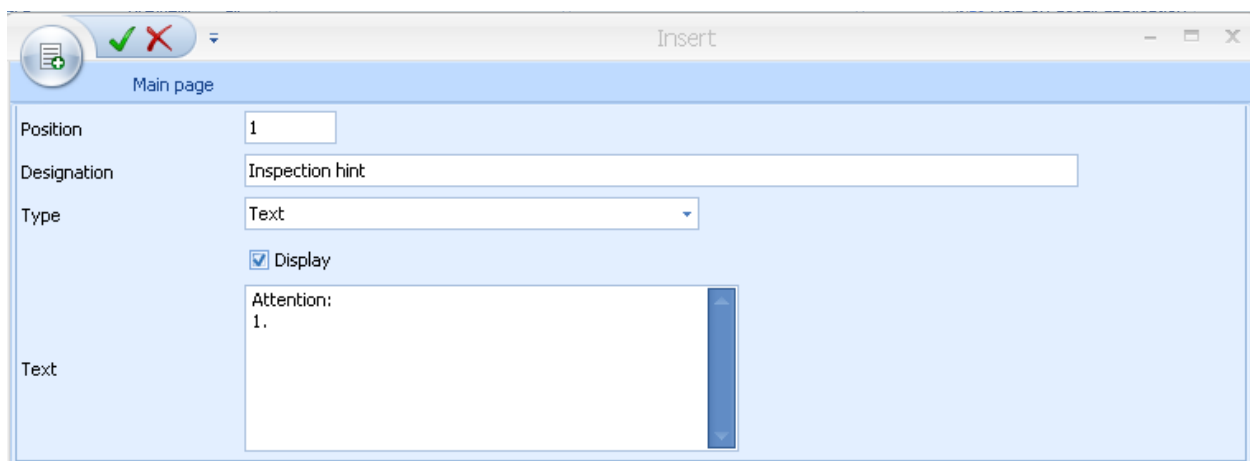
Detail application *Documents*



Characteristics		
Drag a column here to group by that column		
Character...	Characteristic design...	Input type
0001	Total length	manual
0002	Evenness	manual
0003	Length pushbutton	manual
0004	Outer diameter	manual

Documents					
Drag a column here to group by that column					
Position	Designation	Type	File name/address	Display	Display URL/...

If you have activated the tab *Documents*, you can assign an arbitrary number of documents to each characteristic. If this tab is activated, the respective buttons in the toolbar to edit the documents are equally activated.



Insert

Main page

Position: 1

Designation: Inspection hint

Type: Text

☒ Display

Text: Attention: 1.

All formats registered by Windows are available when assigning documents. You can assign simple documents (e.g. written in Word), drawings of any format and videos. You only have to make sure to install a program that is able to display the used format. The appropriate program linked in Windows opens the documents.

The file types "File", "URL", and "Text" are available. If you select the type "file", you can enter the file name including path manually. Select the file type "URL" to access the internet or intranet. Select the file type "text" to directly enter a text.

Note:

The different types of file format "URL" that the shop floor client supports are listed in the respective manual of the shop floor client. It might happen that "https" URL entries are displayed on the MOC, but not on the AIP shop floor client.

You can assign a designation/name to each document. You can also define the list order of the documents. Use the field "position" to define the order (numeric input). Position numbers must be unique in this list. Enable the checkbox *Display* to define that the document is displayed during inspection process.

Speaking of documents, you also have to decide if a document assignment without precise reference to an article is reasonable. Normally, the document assignment depends on the article.

Taskbar Document

In addition to the standard functions, the application also provides the button to show documents.



Show documents

If a document link is stored, click this button to open and show the linked document. However, a program, which can show the linked file type, must be installed on the PC.

12.1 Sampling schemes

The user can select from five sampling schemes in a specified list. Subject to the selected sampling scheme, some additional information has to be defined. It is subject to the subsequent use in the different inspection plan areas (e.g. production, goods receipt, goods issue), if all or only a smaller selection of sampling schemes is available.

Sampling scheme n-c inspection: The sample size is entered in the "sample size" field (= n) and the maximum number of admissible non-conforming units is entered in the "acceptance quantity" (=c) field. The figure "c" is defined as acceptance number. This means: if n = 50 und c = 1, the characteristic and thus the piece is only classified as "fail" if two non-conforming units are identified (with sample size = 50).

Sampling scheme 100% inspection: In general, the sampling scheme *100% inspection* is only used in goods receipt and goods issue. The sample size is calculated from the actual quantity of the inspection requirement and corresponds to it.

Sampling scheme SPC inspection: The sampling scheme "SPC inspection" nearly corresponds to the "n-c" inspection plan. The only difference is that the acceptance limit "c" is not used in this case.

Sampling scheme batch inspection: In the standard configuration, the sampling scheme "batch inspection" only applies to the areas "goods receipt" and "goods issue". The percentage specifying how much percent of the batch is to be checked is entered here. Later in the inspection order characteristic the sample size is calculated from the actual quantity of the inspection requirement and multiplied by the specified percentage.

If you must calculate action limits, you must enter the expected sample size here.

Sampling scheme k-value inspection: With the k value inspection the entered k value is checked against the calculated k value and if this value is violated the sample is rated "fail".

12.2 Control charts for variable characteristics

For variable characteristics, the charts xq, s and R are available.

In statistical quality assurance, production dispersion is used for many calculations. One example is the calculation of capability indices and action limits of a quality control chart. Vice versa, if you have specified a process capability index, you can estimate the production dispersion and calculate the action limits on this basis.

The specifications for the calculation of limit values can be found in the tab "default values chart 1" or "default values chart 2", where values to estimate the production dispersion can be entered. The action and warning limits can be calculated on the basis of these specifications. However, it is also possible to enter the production dispersion directly. The system provides three calculation options.

You first describe the specifications using the xq and s chart. The differences with the R chart are explained in more detail in the sections that follow.

There is often a specification for the process capability index cpk. This specification is reasonable. If the process capability index cpk is respected, you can then produce pieces within the range of tolerance. Based on the specified cpk value, the system calculates internally an estimated value for Sigma, which is entered to the right of the option "Sigma" for information purposes. The estimated basic value that has been calculated is used to calculate the limit values of the xq/s chart. The calculation is performed, once further data has been entered using the *Calculate* button. The calculation method "cpk" is set by default. In addition, there are also the calculation methods "sigma" and "sq/an".

The cpk value of 1,33 ensures that 99.725% of the characteristic values are within the tolerance. However, it is often required that 99.994% of the characteristic values are within the tolerance limit, which corresponds to a cpk value of 1.67.

The calculation method sq/a_n means that an estimate of the standard deviation is calculated from the quotient of the medium standard deviation and a correction factor a_n . This correction factor depends on the sample size, which is identified by the index n . The values for a_n are defined in the system and are requested automatically. This estimate of the standard deviation is best in case that there is no specification of the process capability index and the production dispersion is unknown and thus the specification of the sq -value has still to be corrected by a correction factor. It is also the most efficient method under the given conditions. You must specify the sq -value to calculate the limit values later on. Enter the value in the field on the right hand side of the option sq/a_n . If you click the button *Calculate* later on, the estimate sq/a_n is calculated using the specified sq value. The result is entered on the right hand side of the option *Sigma* for information purposes. This estimate is then the basis for the calculation of action and warning limits of the xq/s chart

The third calculation method requires the specification of a sigma value. In this case it is assumed that sigma is known and consequently the correction factor is not required. Enter the Sigma value to the right of the "sigma" option. In comparison to the previous method, sq/a_n is replaced by sigma. In the majority of cases sigma is not known. Therefore, it is best to use the calculation method using the specified sq value to automatically calculate the estimate sq/a_n for variances in case of doubt.

If you select the "relative deviation from xq " or the "deviation from xq in percent" as "specification to calculate limit values", the input option for "action probability in %" disappears. Instead, you can enter the "deviation from target value". These values and the specified value of xq are then used to calculate the limit values (target value, middle of tolerance, mean value of $\bar{x}$ chart, input).

Further details have to be made in order to identify action and warning limits of the xq chart. You must specify an xq value. The system offers the possibility of setting the xq value equal to the middle of tolerance or the target value or of specifying a value manually. If the process is supposed to be aligned to the mean value, the middle of tolerance should be preferred as xq value.

The action probability must be entered in percent in order to calculate action and warning limits of the xq/s -chart. For this purpose, you must first decide, if you want to use one-sided or two-sided limit values for the calculation. Select one of the two options.

Once you have specified the option 'one-sided' or 'two-sided', enter the action probability in percent. The possible and reasonable action probabilities are defined in the system and only need to be selected from the list. For the xq -chart, the calculation is based on the standard distribution. For example, if 99.725% of the characteristic values must be within the action limits, select the value 99.725. As the specification of a sigma area is commonly used by some users, the corresponding sigma area is displayed with the respective action probability for information purposes.

If warning limits are also to be calculated, an action probability must be entered here. Note: The probability value of the warning limit must be lower than the action limit value.

The sigma area is not displayed in the selection list, since the distribution of χ^2 is used to calculate the limit values of the s-chart. Apart from that, the input is the same as for the xq chart.

If you save the specifications, the limits are calculated – if the *Calculate* checkbox has been enabled before.

As already mentioned, the user can enter the limit values directly without any specified values.

If the R-chart is selected instead of the xq or s-chart, the option sq/a_n is replaced by Rq/d_n in the specifications of the calculation. The estimate of sigma is now calculated using the specified mean range R divided by the correction factor d_n . This correction factor depends on the sample size n. The corresponding values are defined in the system and are selected automatically. Apart from that, the rest is the same as for the xq or s-chart. Note: In case of an R-chart, the calculation of the limit values is not based on a χ^2 distribution, but on a table stored in the system, which is based on standardized ranges.

Notes:

- You can only use the calculation specifications "relative/percentage deviation from xq" if you use an xq control chart. For this reason, the fields of the group "Xq" are only visible, if you have previously selected the xq control chart.
- The user must select the confidence interval (one-sided/two-sided). Usually, you do not define a lower limit for an s-chart. In this case, select "one-sided".
- If only one of the two tolerance limits is available, you cannot use the calculation method "cpk". The calculation formula of the cpk method requires both tolerance limits.

12.3 Control charts for attributive characteristics

The p- and u-charts are available for attributive characteristics.

p identifies the proportion of defective units in the sample and u identifies the failures/defects per unit in the sample. As to the p chart, it is important that each item is either defined as defect-free or defective. If an item has several failures/defects it is only once referred to as defective.

In contrast to the variable characteristics, there are no lower limit values. Furthermore, it is normally not necessary to state the values UTL, LTL and target value.

It is necessary to enter a pq or uq value in percent for the automated calculation of specifications. This can be done in the default values tab.

If you save the specifications, the limits are calculated – if the *Calculate* checkbox has been enabled before.

Calculation is respectively based on normal distribution. The value 99,725 has to be selected if, e.g. 99,725% of the characteristic values are supposed to lie below the upper action limit. As the specification of a sigma area is commonly used by some users, the corresponding sigma area is displayed with the respective action probability for information purposes.

12.4 Calculation of formulas

If you store a formula, you can automatically calculate measured values by way of measured values or statistical values of other characteristics that have been inspected before.

If the extension [QMSingleValue.FormulaArguments](#) is enabled, you have the possibility to use extensive arguments to calculate the single value you want to collect. In addition, you have more possibilities to access specification values and values of inspection results of other characteristics. For more details, refer to the section "".

If this extension is not enabled, you can only calculate characteristics using the inspection results of other characteristics that have already been entered. Find details in the section "Calculation via reference to other inspection results".

12.4.1 Operators, functions and constants

The following operators, functions and constants for calculating measured values are supported:

Functions	
abs(x)	Calculates the absolute value
atan(x)	Calculates the arc tangent
cosh(x)	Calculates the hyperbolic cosine
float(x)	Converts the value into a floating point number
sqrt(x)	Calculates the square root
acos(x)	Calculates the arc cosine
atan2(y,x)	Calculates the arc tangent of y/x
exp(x)	Calculates the exponential value
log(x)	Calculates the natural logarithm
sin(x)	Calculates the sine
tan(x)	Calculates the tangent
asin(x)	Calculates the arc sine
cos(x)	Calculates the cosine
int(x)	Converts the value into an integer
log10(x)	Calculates the common logarithm
round(x)	Rounds to integer value
round(x,y)	Rounds the value x to y decimal places
sinh(x)	Calculates the hyperbolic sine
tanh(x)	Calculates the hyperbolic tangent
trunc(x)	Reduces the value x to an integer value
trunc(x,y)	Reduces the value x to y decimal places
Operators	
x + y	Addition
x - y	Subtraction
x / y	Division
x * y	Multiplication
x ** y	Calculates x to the power of y
Constants	
pi	3.141592654
e	2.718281828

If constant numeric values are used in formulas, you must be careful not to use thousand separators. If these constants are floating point numbers, be careful to use a dot as decimal separator instead of a comma.

12.4.2 Formulas referring to other inspection results

Formulas including a reference to other inspection results are always calculated when an inspection result referenced in the formula is created, changed or deleted.

For these characteristics, you must first specify the level of the formula calculation. The following types are available:

- **V** – Calculation on the level of single values (Value).
For each single value of the characteristics involved, one single value is generated for the calculated characteristic.
- **S** - Calculation on the level of samples (Sample).
For each sample of the characteristics involved, exactly one single value is generated for the calculated characteristic.
- **C** - Calculation on the level of characteristics (Criteria).
Exactly one single value is generated for the calculated characteristic (with respect to the overall statistic of all characteristics involved)

The actual formula follows this identifier (see previous chapter).

The following syntax applies for the variables identifying the single values or statistical values of the order characteristics involved [x:y:z].

The x parameter identifies the statistical value to be used. The available values are listed below. Please bear in mind that the calculation level might cause restrictions.

- **X** – Single value
(is only available for calculations on the level of single values)
- **AVG** – Mean value
(is only available for calculations on the level of samples or characteristics)
- **MIN** – Minimum
(is only available for calculations on the level of samples or characteristics)
- **MAX** – Maximum
(is only available for calculations on the level of samples or characteristics)

- SUMX – Sum of single values
(is only available for calculations on the level of samples or characteristics)
- R – Range
(is only available for calculations on the level of samples or characteristics)
- S – Standard deviation
(is only available for calculations on the level of samples or characteristics)
- N – Sample size
(is only available for calculations on the level of samples or characteristics)
- M – Number of samples
(is only available for calculations on the level of characteristics)

The y parameter describes how the corresponding characteristic is supposed to be identified. The following possibilities are available:

- SENO – identification via the OP sequence of the characteristic (serial number)
- INCR – Identification via the characteristic number (inspection criteria)
If the characteristic number is not unique within the inspection requirement, it is not predictable which one of the applicable characteristics is used at the time of calculation.

The parameter z identifies the characteristic using the field content defined by parameter y . Either the OP sequence or the characteristic number of the calculation source is entered in this field. If the characteristic number includes a space character, it should be replaced by an underscore within the formula.

Example 1:

A new characteristic is calculated from the single values of the characteristic assigned to the number "LENGTH"/"LAENGE" divided by 2.5. A corresponding single value is supposed to be calculated for each single value of the source characteristic (calculation on the level of single values).

→ Formula: $V: [X:INCR:LAENGE] / 2.5$

Example 2:

The characteristic "surface" results from the product of the characteristics with the characteristic number "LENGTH"/"LAENGE" and "WIDTH_TOTAL"/"BREITE_GES". A single value of the characteristic "surface" is supposed to be calculated for each single value of both source characteristics (calculation on the level of single values).

→ Formula: V: [X:INCR:LAENGE] * [X:INCR:BREITE_GES]

Example 3:

The characteristic "maximum margin width" results from the subtraction of the minimum of the characteristic "inside diameter" (OP sequence 10) from the maximum of the characteristic "outside diameter" (OP sequence 20). A single value of the characteristic "maximum margin width" is supposed to be calculated for each sample of both source characteristics (calculation on the level of samples).

→ Formula: S: [MAX:SENO:20] - [MIN:SENO:10]

For the calculation of formulas including references to other inspection results, it is allowed to calculate new formula characteristics that are based on calculated formula characteristics. However, this nesting may not have more than 10 references one below the other. Furthermore, double concatenations must not be created (Example: characteristic A is calculated from characteristic B and characteristic C; characteristic C is calculated from characteristic A).

12.4.3 Extended formulas

The extended formulas provide the following advantages compared to the formulas including references to other inspection results:

- You can enter arguments for these characteristics that are used to calculate the measured value. In most cases, you do not need to use other "source characteristics".
- On saving the inspection result, the measured value is calculated and is immediately available. You do not need to refresh the measured values in the AIP to see the measured values.
- For the calculation, you can optionally use single values or sample or characteristic statistics of other characteristics. You may combine these in any way.
- You can use variables for the target value, the upper and the lower tolerance limit of the current characteristic or of other characteristics in the formula.

For this reason, you should primarily use the extended formulas.

The following syntax applies for the variables identifying the single values, statistical or specification values of the order characteristics involved $[x:y:z]$.

The x parameter identifies the statistical value to be used. The available values are listed below. Note: Depending on the respective shop floor client used, it is possible that not all 10 argument fields are available.

- VAR1 – Argument 1 of the inspection result of the own inspection step characteristic
- VAR2 – Argument 2 of the inspection result of the own inspection step characteristic
- VAR3 – Argument 3 of the inspection result of the own inspection step characteristic
- VAR4 – Argument 4 of the inspection result of the own inspection step characteristic
- VAR5 – Argument 5 of the inspection result of the own inspection step characteristic
- VAR6 – Argument 6 of the inspection result of the own inspection step characteristic
- VAR7 – Argument 7 of the inspection result of the own inspection step characteristic
- VAR8 – Argument 8 of the inspection result of the own inspection step characteristic
- VAR9 – Argument 9 of the inspection result of the own inspection step characteristic
- VAR10 – Argument 10 of the inspection result of the own inspection step characteristic
- X – Single value of another characteristic
- AVG – Mean value of the sample of another characteristic
- MIN – Minimum of the sample of another characteristic
- MIN – Maximum of the sample of another characteristic
- SUMX – Sum of the single values of the sample of another characteristic
- R – Range of the sample of another characteristic

- S – Standard deviation of the sample of another characteristic
- SREL – Relative standard deviation of the sample of another characteristic
- N – Sample size of another characteristic
- AVG_ALL – Mean value of all samples of another inspection step characteristic
- MIN_ALL – Minimum of all samples of another inspection step characteristic
- MAX_ALL – Maximum of all samples of another inspection step characteristic
- SUMX_ALL – Sum of the single values of all samples of another inspection step characteristic
- R_ALL – Range of all samples of another inspection step characteristic
- S_ALL – Standard deviation of all samples of another inspection step characteristic
- N_ALL – Total sample size of all samples of another inspection step characteristic
- M_ALL – Number of samples of another inspection step characteristic
- TV – Target value of an inspection step characteristic
- UTL – Upper tolerance limit of an inspection step characteristic
- LTL – Lower tolerance limit of an inspection step characteristic

The *y* parameter describes how the corresponding characteristic is supposed to be identified. The following possibilities are available:

- SENO – identification via the OP sequence of the characteristic (serial number)
- INCR – Identification via the characteristic number (inspection criteria)

If the characteristic number is not unique within the inspection requirement, it is not predictable which one of the applicable characteristics is used at the time of calculation.



The characteristic number must not include any special characters. A minus sign "-" is not permitted, for example.

- **SELF** – Identification of the own calculated characteristic

The characteristic that is to be calculated identifies itself. Only in this case, the parameter *z* is not required.

Note: You may only use the identification of the own characteristic for the argument fields and for the target value and the tolerance limits.

The parameter *z* identifies the characteristic using the field content defined by parameter *y*. Either the OP sequence or the characteristic number of the calculation source is entered in this field. If the characteristic number includes a space character, it should be replaced by an underscore within the formula.

Example 1:

The measured value of the current characteristic is calculated from the sum of the argument fields 1 to 4.

→ Formula: `[VAR1:SELF] + [VAR2:SELF] + [VAR3:SELF] + [VAR4:SELF]`

Example 2:

The characteristic is the result of the product of the maximum measurements of the inspection step characteristics with the characteristic numbers 'LAENGE' and 'BREITE_GES' ('LENGTH' and 'WIDTH_TOTAL').

→ Formula: `[MAX_ALL:INCR:LAENGE] * [MAX_ALL:INCR:BREITE_GES]`

Example 3:

The measured value is calculated from the sum of the following three summands:

- Content of argument field 1
- Middle of the tolerance of the current characteristic
- Sample mean value of the characteristic with OP sequence 10

→ Formula: `[VAR1:SELF] + (([UTL:SELF] + [LTL:SELF]) / 2) + [AVG:SENO:10]`

Note the following when using extended formulas:

- Contrary to the formulas including references to other inspection results, the measured values are not calculated when the "source characteristics" are changed. The measured values are only calculated, if the inspection result of the respective calculated characteristic is explicitly collected or changed (e.g. via the argument fields).
- When the inspection result is saved, the system must be able to identify valid values for all variables used in the formula (single values, sample or characteristic statistics, specification values of other characteristics, all used arguments).
Otherwise, an error message occurs and the inspection result is not saved.
- You cannot directly edit the calculated measured value. The measured value is always the result of a calculation.
- If you use the parameter [X:...], the respective single values of other characteristics are searched for using the absolute single value and sample number. For the current characteristic, the parameter [X:...] is not available.
- If you use the statistical parameters [MAX:...], [MIN:...], [AVG:...], [SUMX:...], [R:...], [S:...], [SREL:...], or [N:...], the respective statistical values are searched for using the absolute sample number. Here, you cannot use statistical parameters of the own characteristic.
- If you want to use the statistical parameters of the complete characteristic using the parameters [MAX_ALL:...], [MIN_ALL:...], [AVG_ALL:...], [SUMX_ALL:...], [R_ALL:...], [S_ALL:...], [M_ALL:...], or [N_ALL:...], only the data of other characteristics is available (not the data of the own characteristic).
- Via customization, extensions can be made available to obtain any variables in the syntax [VAR:<Object>:<Identifier>].
- You cannot use characteristics that include extended formulas as sources to calculate formulas including references to other inspection results. But you can use these characteristics for other characteristics with extended formulas.

12.4.4 General notes on calculated characteristics

If unknown variables are used within a formula (faulty parameters x and/or y), the escalation CPAUMW.CALCULATED_CRITERIAS_GET_VARIABLE_VALUE is triggered.

If problems occur on assigning an identified value to a variable of the formula, the escalation CPAUMW.CALCULATED_CRITERIAS_SET_VARIABLE is triggered.

Both actions described require the escalation management license.

Tool numbers, machine numbers, cavity numbers or similar information are not stored for the calculated single values.

To transfer a corresponding number (batch number, sample number, serial number, etc.), all source samples of the calculation must be assigned the same number. If there is no number that is assigned to all source samples, you cannot assign a number to the calculated sample. If several numbers are found that have been assigned to all source samples, only the first number found is assigned to the calculated sample.

This function only applies to numbers, which have been assigned on sample level.

12.5 Last off inspection

As part of the function extension for the in-production inspection, the function of the last off inspection is available. For this function, you must have created the CAQ system option 1222 manually as a precondition. For details, refer to the procedure document "[Configuration_OM_Options.pdf](#)". The documentation of the CAQ system option specifies which characteristic user fields must be created (master data characteristic, inspection plan characteristic and inspection step characteristic), so that you can specify a characteristic for a last off inspection.

To specify a characteristic for a last off inspection, the user field of the last off inspection must have a content.



The function "Last off inspection" is not offline capable. And you cannot use the last off inspection with operations that are specified as *Inspection OP* via processing code.

If the operation is logged off or interrupted in offline mode, the buffered activities/postings are processed one after the other when the online mode is restored. This has the effect that the operation is logged off or interrupted although the last off inspection is missing.



The processing code generally defines if any check for a defined last off inspection is performed at all during logoff/interruption. If the processing code in tab *Quality* is set to *Inspection OP*, no check and no last off inspection is performed.

If an inspection step has been "logged on" with the logon of an operation on the AIP, the system proceeds as follows for the last off inspection when the operation is logged off or interrupted.

1. The system checks if an inspection point with cause for creation *Last off inspection* exists for this inspection step at the workplace in question. It does not matter if the inspection point is completed or not. If the check is also performed with an interruption, the system checks if an inspection point with cause of creation *Last off inspection* has been created since the last logon. If an inspection point is found, the operation is interrupted or logged off.
2. If no inspection point is found in item 1, the system checks if the relevant inspection step logged on includes characteristics with the inspection due date *Last off inspection*. If this is not the case, the operation is logged off or interrupted.

3. If the processes described in item 1 and 2 have the result that an inspection point must be created with the cause of creation *Last off inspection*, the following message is shown:

Last off inspection is missing!

Enforce posting using the option "posting required"?

Using the option *Posting required*, you can perform the logoff/interruption without having made a last off inspection. Use the CAQ system option 1154 to activate the logoff/interruption using the option *Posting required*.

4. If the operation is not logged off or interrupted because of the missing inspection point with the cause of creation *Last off inspection*, you must go to the inspection to create an inspection point with the cause of creation "Last off inspection". You can use the option *Last off inspection* in the inspection list on the level *Inspection step* to create an inspection point with cause of creation *Last off inspection*.

If you try to log off or interrupt the operation that includes an inspection point with cause of creation *Last off inspection* that has not been completed, the standard processes apply. This means that the operation can optionally be logged off or interrupted although the last off inspection has not been performed/completed using the option "Posting required".

13 Composition

Summary

Menu	Material management → Composition → Composition
Transaction code	comp
Function authorization	comp

Usage

You use the "composition" application to control planning, monitoring and execution of the melting process. This application includes the following functions:

- Generation of charging orders
- Performance of composition planning based on existing and advised stocks as well as the planned bottom sump
- Reservation of the planned material for the planned melt
- Generation of a charging list
- Release for material provision (perform charging)
- Execution of composition based on the provided materials and the actual bottom sump
- Checking of the theoretical analysis
- Saving of theoretical analyses
- Display of the target analysis including tolerances and restrictions
- [*Execution of charging and sampling at the terminal*](#)
- Comparison of actual analysis and target analysis by identifying the elements using traffic light colors.
- Performance of re-composition

The configuration of composition functions are described in the relevant [procedure](#).

Selection criteria

The following selection criteria are available in the application:

Category "charging order"

The selection criteria of this category have been designed to restrict data in the detail application "charging orders".

Order

Charging order

Order status

Order status of the charging order

Finished article

Finished article of the charging order

Machine

Planned melting aggregate

"Dates" category

The selection criteria of this category have been designed to restrict data in the detail application "charging orders".

Basic start date / basic end date

Temporal restriction using basic dates

Scheduled start/scheduled end

Temporal restriction by scheduling

Category "input material"

The selection criteria of this category have been designed to restrict data in the detail application "inventory of permitted materials".

Material

Input material.

Material designation

Designation of input material.

Material buffer

Material buffer to identify the material

Material type

Material type for classification

Category "options"

The selection criteria of this category have been designed to restrict data in the detail applications "inventory of permitted materials" as well as "charging orders".

Show additional material only

If this option is enabled, the detail application "inventory of permitted materials" will only show materials configured as "additional materials" for the output material (finished article of the charging order) that is to be produced by the charging order.

Show permitted material only

If this option is enabled, the detail application "inventory of permitted materials" will only show materials configured as "permitted input materials" for the material that is to be produced by the charging order.

Absolute values for materials

Displays quantities as absolute values or in percent in the detail application "inventory of permitted materials".

Absolute values for order materials

Displays quantities as absolute values or in percent in the detail applications "selected materials", "composition recipe" and "samples".

Detail application "charging orders"

The detail application "charging orders" shows the list of charging orders.

Charging order

The charging order is indicated with planned order, finished article as well as target quantity and unit.

Planning

Shows the planned dates as well as the melting aggregate

Status

Shows the order status as well as composition status. The composition status results from the last sample taken for this charging order. If this sample is ok ("pass") the composition status is green; if it is not ok ("fail") the status is red.

Price

Shows the average price per kg in € for the material to be produced, the current price per kg in € as well as the total price in €.

Miscellaneous

Displays the scrap rate relating to the selected materials and their properties configured in the material master. The column is highlighted in red if at least one formula for checking the scrap rate is not met. It is highlighted green if all defined formulas are met.

Shows the weight and max. capacity of the melting aggregates used in the order.

Detail application "inventory of permitted materials"

Subject to the entered selection criteria, the list only shows materials that are permitted for the output material or materials that are permitted and have been defined as additional material.

Only will material be taken into account that has been assigned to a casting buffer and to a material type that has been defined as anonymous material (inventory management).

Available quantity

Remaining quantity of a material that is in the batch status "free" and that has not been reserved.

Advised quantity

Remaining quantity of a material that is in the batch status "advised".

Reserved quantity

Remaining quantity of a material that is in the batch status "free" and that has been reserved.

Detail application "selected materials"

This detail application shows the assigned materials.

Input type

Current bottom sump: material that can currently be found in the output buffer of the machine (melting aggregate). The material is determined by the function "adjust bottom sump".

Planned bottom sump material that will be available in the machine's output buffer. The material is determined by the function "adjust bottom sump".

Reserved: material which has been assigned manually using the function "assign material".

BOM item

Planned bottom sump material or the current bottom sump is indicated by the BOM item "0". The material is assigned the BOM item "1" at first. Once charging has been performed and confirmed at the AIP terminal, all materials pertaining to this BOM item are grayed out and the BOM item "1" is blocked. Further materials are now assigned to the next BOM item. Consequently, the current BOM item is blocked every time charging has been performed and the BOM item is increased by one if further materials are added.

Material, designation

Material

Target quantity

Target quantity planned for the production of the finished material. This quantity can be changed by double clicking or the shortcut <Ctrl + Q>. All calculations (formulas, samples: theoretical analysis, charging orders: composition status and scrap rate) are based on the target quantity.

Reserved quantity

Actually reserved quantity that has been reserved using the function "perform reservation" within the batch stock.

Consumed quantity

Actual quantity that has already been consumed and placed in the melting aggregate.

Editor, editing time

The user who has performed the assignment is entered here. The editing time is updated, once the charged material has been confirmed at the AIP terminal.

Detail application "composition recipe"

This detail application shows the composition recipe for the material (finished article) of the selected charging order. The components/elements are indicated with target quantity ("composition recipe" row) as well as with the upper and lower tolerance limits ("upper/lower tolerance limit" rows).

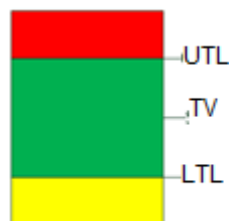
Detail application "formulas"

This detail application shows the composition restrictions for the finished material and the results of the corresponding formula. If the formula is true, the fields are highlighted in green; in any other case they are red.

Formulas are calculated based on the latest sample, i.e. on the current composition of the material (from the analysis process determining the make-up of the sample) and the quantity that is actually entered with this sample at the AIP terminal. All materials assigned after sampling and that have not yet been charged are considered with the relevant target composition and target quantity. All materials that have been charged and confirmed at the AIP terminal after taking the sample are considered with the target composition and the consumed quantity. The calculation is based on all assigned materials, provided that a sample has not yet been taken.

Detail application "Samples"

This detail application shows the samples with the respective sample ID, date and time as well as status (PPKT_NIO=inspection point failed, PPKT_IO=inspection point pass). The share of each element is shown and evaluated (green = value within tolerance limits, yellow = value below the lower tolerance limit, red = value above the upper tolerance limit).



A theoretical analysis is shown additionally referring to the last sample (provided that samples have already been entered) and its current composition as well as to the materials assigned subsequently that might have been charged already - just as it is also the case for the calculation of formulas.

Once the charging order has been released, the theoretical analysis is saved as the "release analysis" ("theoretical analysis for the release").

Toolbar

Start composition



Generate charging order

Provided that no charging order exists for a melting order, this function can be used to generate one.



Start composition for charging order

In case a charging order exists or has been generated, composition is started for the charging order by using the function of the same name. Data of the charging order and of the assigned melting order are shown.

Material reservations



Assign material

Inserts the material or materials selected in "inventory of permitted materials" with the quantity 0 kg for the selected charging order in the "list of charging orders". The material cannot be assigned, provided that the BOM item has already been assigned this material and has not yet been blocked (by confirming the charging process at the AIP terminal).

Change quantity

Changes the target quantity of a selected material. This editing dialog cannot be opened via the quick launch bar. It can only be opened by double clicking a row of the detail application "selected materials" or the shortcut <Ctrl+Q>

The quantity can only be changed if the new quantity is greater than or equal to the quantity that has already been consumed/used. If this is not the case, the user will be informed by an error message.

The quantity cannot be changed if the BOM item is blocked, i.e. the charging process has been confirmed for all materials of this current BOM item at the AIP terminal.



Perform reservation / perform reservation (including material)

This function reserves materials for the selected charging order. Consequently, the required quantities of materials are reserved for this charging order (if available in sufficient quantities) and cannot be used for any other charging order. The reservation applies for all unblocked entries of the order's component list by using the function "perform reservation". As an alternative, it can only be performed for the transferred material by using the function "perform reservation (including material)".

**Cancel reservation / cancel reservation (including material)**

The reservation of materials is canceled for all unblocked entries of the order's component list using the function "cancel reservation" or only for the selected material using the function "cancel reservation (including material)". A reservation can only be canceled completely, provided that material has not yet been consumed.

**Adjust bottom sump**

This function transfers the material of the batch that is currently within the furnace (current bottom sump) to the currently active composition. In this case, the current bottom sump quantity is also taken over. The theoretical values (planned bottom sump) assumed up to now and resulting from the sequence of orders are overwritten by current information. The bottom sump can no longer be adjusted, once the first charging process has been performed and confirmed at the AIP terminal.

**Copy recipe**

This function copies the recipe from one charging order to the other. The reservation can be performed afterwards.

Release

**Release of charging order**

Once the charging order has been released and the materials have been selected, the order is available for execution in production. The "theoretic analysis" presented in the samples is saved as the "release analysis".

**Charging list**

The charging list can be printed for the released charging order. This list includes the melting order, charging order and a list of required materials.

Further functions

**Reuse**

If the required material cannot be achieved by adding material (as the melting aggregate's capacity would, for example, be exceeded), the charging order can be assigned to another or new melting order.

**Melting report**

The melting report outputs all pieces of information referring to the melting process:

- Charging order
- Assigned material and quantity in kg
- Composition recipe
- Current analysis including samples and theoretical analysis

14 Composition - AIP

Definition (source: <http://de.wikipedia.org/wiki/Gattierung>)

In foundry, composition (optimization of composition) means the make-up of foundry material that is melted down in the melting furnace. Composition is necessary to achieve an as exact chemical make-up of the cast material as possible without having to add many ingredients. The result is kept in a composition list acting as a bill of material.

However, it is required to have detailed knowledge about the chemical analysis of the raw materials, i.e. normally pig iron, steel scrap and recycled material. Based on this analysis, it is calculated in which proportion they have to be put into the furnace to achieve the required characteristics of the material.

Composition has to be optimized if the charge make-up is to be performed in a cost-effective manner. To do so, the required quantities of the raw materials, such as steel scrap or pig iron are calculated using mathematical procedures in order to achieve the required characteristics of the material by combining the raw materials as economically as possible. Therefore, an exact chemical analysis of raw materials is required. The mathematical calculation is performed using specialized computer software for the optimization of composition.

14.1 Perform charging

The dialog "perform charging" shows the current list of materials the furnace is fed with. The list includes the planned, reserved and already fed quantities as well as the required remaining quantities.

The quantities can be edited here.

Perform charging

Workplace 22222222 Casting

Order 2222222205 Operation 0100

Material	Materialdesc	Target qty	Unit	Reserved qt	Loaded qt	Materialbuffer	Remaining qt	Input buffer
8260	8260 // Al 97,95 Si 0,1	9000,000	kg	9000,000	9000,000	22222222E		9000,000
Al	aluminum (pure)	100,000	kg	100,000	50,000	22222222E		50,000
			kg			2222-03	50,000	0,000
Fe	iron (pure)	20,000	kg	20,000	15,000	22222222E		15,000
			kg			2222-01	5,000	0,000

Material: 8260

Quantity: KG

Material buffer: 22222222E

Staff badge number:

Close Post

mpcv CTAIP V2.0.2.48 Server: win2008-6.1 TNR:222 22.01.13 09:13:08

Figure: Perform charging – C_CHPF

Data from the selected row are entered in the below input fields

The button "post" prompts posting of the material in the machine's input buffer. Then the list is reloaded. For this purpose, the terminal needs to be connected online with the HYDRA server.

The dialog remains opened until the button "cancel" is clicked.

The list does not show the material that has already been confirmed by the dialog "confirm charging". Consequently, the list of the dialog "perform charging" is empty directly after finishing the charging process. Unplanned material may also be added to the dialog "perform charging". This material is then shown in the list.

14.2 Confirm charging

The current list of materials is shown again to confirm charging. Besides the badge number, no further input is required.

Confirm charging

Workplace 22222222 Casting
 Order 2222222205 Operation 0100

Material	Materialdesc	Target qty. Unit	Reserved qt	Loaded qt	Materialbuffer	Remaining qt	Input buffer
8260	8260 // Al 97,95 Si 0,1	9000,000 kg	9000,000	9000,000	22222222E	9000,000	9000,000
Al	aluminum (pure)	100,000 kg	100,000	50,000	22222222E	50,000	50,000
		kg			2222-03	50,000	0,000
Fe	iron (pure)	20,000 kg	20,000	15,000	22222222E	15,000	15,000
		kg			2222-01	5,000	0,000

Staff badge number

CTAIP V2.0.2.48 Server: win2008-6.1 TNR:222 22.01.13 09:12:20

Figure: Confirm charging – C_CHCF

Once confirmed, the materials from the input buffer are posted onto the machine's output buffer. The reservation of remaining quantities of materials reserved for the charging order is cancelled.

14.4 Take sample

A sample is taken to check the actual composition of the melt.

Take sample	
Order	MPDV400003
Material	4711-2
Quantity	50.000
Staff badge number	
Cancel Take sample	

Figure: Take sample – C_CHTS

Using this dialog, a sample number is assigned by the HYDRA server and returned as the result to the terminal. A label that includes the sample number is printed. Consequently, the relevant label has to be configured and assigned to the dialog C_CHTS.

The “quantity” field is assigned the quantity of the output buffer by default. This requires, however, that the terminal is connected online with the HYDRA server.

14.5 Cast

Casting means to withdraw a (partial) quantity from the melt.

Order	MPDV400003
Operation	0020
Material	4711-2
Mat. descript.	MPDV Dose
Quantity	50.000
Material buffer	
Batch	PRNZG2Q112
Staff badge number	

Cancel Cast

Figure: Casting – C_CHCA

The withdrawn quantity and the target buffer to which this quantity is posted are entered in this dialog. The terminal determines the batch number automatically, provided that the "automatic generation of the batch number" has been enabled in the "MPL" tab of the workplace configuration.

The "quantity" field is assigned the quantity of the output buffer by default. This requires, however, that the terminal is connected online with the HYDRA server.

14.6 Implementation / configuration

The dialogs C_CHPF, C_CHCF, C_CHTS and C_CHCA have to be available at the terminal. The buttons in ctaipbut.ini are configured by using the same IDs:

```
1=C_CHPF,R,perform charging
2=C_CHCF,R,confirm charging
3=C_CHTS,R,take sample
4=C_CHCA,R,cast
```

The section [charge list] is required in the layout configuration of ctaipplay.ini:

```
[charge list]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite

MAT_VIS=C25,80,L
MATBEZ_VIS=C30,115,L
SOLL_MENGE_VIS=N12.3,78,R,target quantity
EINH=C3,40,L
RES_MENGE_VIS=N12.3,78,R,remaining quantity
VERBR_MENGE_VIS=N12.3,78,R,delivered quantity
MATPUF=C30,80,L
MATPUF_MENGE=N12.3,78,R,remaining quantity
EMAT_MENGE_VIS=N12.3,78,R,input buffer
```

Further configuration details are described in the relevant [procedure](#) for composition functions.